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Programming for Artificial Intelligence and Data Science (CO3135)

Assignment Submission

House Price Prediction

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1 Group Information & Member Roles

The group consists of five members working on house price prediction for the Vietnamese real estate market. Responsibilities are organised into dataset preparation and exploratory data analysis, machine learning, and deep learning. Table 1 lists the individual contributions.

Table 1: Group members and assigned responsibilities.

Member	Area	Responsibilities
Nguyễn Mạnh Quốc Khánh	Dataset and EDA	Dataset selection and preparation; schema and missing-value analysis; duplicate and outlier handling; exploratory analysis of price, area, location, and property type; preparation of the dataset and EDA report.
Võ Nhật Tân	Machine learning	Structured Ridge regression, Random Forest, LightGBM, and CatBoost; model-specific preprocessing, training, validation, and interpretation of the assigned models.
Phan Ngọc Lan Chi	Machine learning	XGBoost, text-only TF-IDF Ridge, joint text-and-structured Ridge, and LightGBM with a text-derived prediction feature; text representation, model training, validation, and interpretation of the assigned models.
Ngân Phạm	Deep learning	Tabular multilayer perceptron with categorical embeddings; numerical preprocessing, embedding construction, network implementation, optimisation, and early stopping.
Minh Thiện	Deep learning	BamiBERT text regression; text preparation, tokenisation, target standardisation, pretrained-model fine-tuning, and checkpoint evaluation.

Shared machine learning responsibilities. Võ Nhật Tân and Phan Ngọc Lan Chi jointly establish the constant-median and local median-price-per-square-metre baselines, prepare the shared feature representation and chronological evaluation protocol, compare the machine learning models, and write the machine learning report. Each member is assigned four learned model configurations. The text-only TF-IDF Ridge model also supplies the prediction feature used by the text-enhanced LightGBM.

Shared deep learning responsibilities. Ngân Phạm and Minh Thiện are each responsible for one principal neural model. They jointly compare the deep learning results, implement and evaluate the prediction blends combining structured and text-based models, and write the deep learning report.

2 Chosen Data & Justification

The chosen dataset provides a large number of real-estate listings with a clear numeric target and a wide range of predictors, which suits the house price prediction task for the Vietnamese market.

Dataset *Tinix Vietnam Real Estate Listings 2025–2026* [Tin26]

Overview According to the published dataset card, the collection contains exactly 3,500,744 sale and rental listings published between June 2025 and March 2026, after a strict property-type filter, and covers all 63 provinces and cities of Vietnam. The listings are distributed as ten Parquet shards, are written primarily in Vietnamese, and are released under the CC BY-NC 4.0 licence for non-commercial academic use.

Justification The dataset contains over 3.5 million records, which supports stable exploratory analysis and supervised learning across regions and property types. Its schema separates a numeric target (`price`) from predictors describing location, property category, size and layout, orientation, textual content, and listing time. National coverage makes it possible to study regional price patterns and to train models that generalise beyond a single city, and the publication timestamp allows chronological train/validation/test splits, which are preferable to random splits for a market that changes over time. The publisher also lists real-estate price prediction among the intended use cases.

Limitations The published definitions indicate that `price` may be undisclosed (`null`), that some structural fields are often incomplete for particular property types, and that the collection includes both sale and rental listings, with no variable that distinguishes them. These issues are examined in Section 4, where an analytical subset of priced listings is constructed for subsequent analysis and model training.

Table 2 reproduces the data schema from the published dataset card. The descriptions are taken from that page and are not reinterpreted here.

Table 2: Feature meanings from the published dataset card [Tin26].

Column	Type	Description (published)
`name`	string	Listing title (HTML stripped, phone numbers redacted)
`description`	string	Full listing description (HTML stripped, phone numbers redacted)
`property_type_name`	string	Property category (e.g., Căn hộ chung cư, Nhà, Đất, ...)
`province_name`	string	Province / City name (63 provinces)
`district_name`	string	District (Quận / Huyện)
`ward_name`	string	Ward (Phường / Xã)
`street_name`	string	Street name
`project_name`	string	Real estate development project name (if applicable)
`price`	float64	Listed price in VND (Vietnamese Dong). `null` if not disclosed.
`area`	float64	Total floor/land area in m^2
`floor_count`	float64	Number of floors in the building
`frontage_width`	float64	Frontage width in meters
`house_depth`	float64	Depth of the property in meters
`road_width`	float64	Width of the road in front of the property (meters)
`bedroom_count`	float64	Number of bedrooms
`bathroom_count`	float64	Number of bathrooms
`house_direction`	string	Cardinal facing direction (Đông, Tây, Nam, Bắc, Đông-Bắc, ...)
`balcony_direction`	string	Cardinal facing direction of the balcony
`published_at`	string	ISO 8601 datetime the listing was published

3 Work Progress Overview

The project developed through weekly milestones between 31 August and 11 November 2026. Table 3 summarises the deliverables reached at each stage, the main difficulties, and the corresponding adjustments. Khánh led dataset preparation and EDA; Tân and Chi developed the machine learning models; Ngân and Thiện developed the neural models and prediction blends.

Table 3: Weekly progress, difficulties, and responses.

Week	Progress reached	Difficulty and response
1 31 Aug.–6 Sep.	Agreed on asking-price prediction, selected the Vietnamese real estate dataset, reviewed the available attributes, and divided the work among five members.	Listings include both sale and rental advertisements without a reliable transaction-type label. The group defined the target as advertised asking price and retained this ambiguity as a limitation.
2 7–13 Sep.	Completed initial loading, schema checks, data-type inspection, and missing-value summaries. Established a reusable local data copy for subsequent analysis.	The dataset contains approximately 3.5 million listings, making repeated loading costly. Local Parquet storage and categorical conversion reduced repeated data-access and memory overhead.
3 14–20 Sep.	Completed duplicate detection and checks for invalid price and area values. Produced missing-value summaries by property type.	Many structural attributes are absent, and missingness differs by property type. The group avoided dropping every incomplete listing and retained missingness for model-specific treatment.
4 21–27 Sep.	Completed the main price and area analyses, reviewed extreme observations, and established the cleaned modelling population of 2,397,001 listings.	Extreme prices, implausible dimensions, and invalid timestamps affected modelling suitability. Price, area, and price-per-area restrictions were applied; invalid optional dimensions were set to missing.
5 28 Sep.–4 Oct.	Completed hierarchical location keys, categorical missing-value handling, text preparation, and the shared chronological 70/15/15 split.	Repeated district names could conflate locations, while advertisements could disclose the target price. Parent-location prefixes and monetary-clause/digit removal addressed these issues, with residual text leakage retained as a limitation.
6 5–11 Oct.	Established constant-median and local price-per-square-metre baselines. Completed structured Ridge training and the three-value regularisation search.	Linear fitting required consistent treatment of skewed numeric inputs, missing values, and numerous categorical levels. Training-fitted log transforms, imputation, scaling, and rare-category pooling formed a common pipeline.
7 12–18 Oct.	Completed the Random Forest depth/feature-fraction comparison and the LightGBM, CatBoost, and XGBoost training configurations. Prepared validation metrics and feature-importance diagnostics.	Tree experiments on the full training population were computationally demanding. A 10% bootstrap fraction per forest tree, parallel CPU execution, and validation-based stopping controls limited the training workload.
8 19–25 Oct.	Completed TF-IDF features, joint text-and-structured Ridge, and the text-prediction input for LightGBM. Completed the tabular MLP architecture and its preprocessing pipeline.	Dense text matrices would be costly, and in-sample text predictions could mislead the downstream model. Sparse matrices and five-fold out-of-fold Ridge predictions were used; categorical embeddings reduced the neural input expansion.

Continued on the next page.

Table 3 continued.

Week	Progress reached	Difficulty and response
9 26 Oct.–1 Nov.	Developed the MLP optimisation and early-stopping procedure. Prepared BamiBERT tokenisation, standardised log-price targets, and GPU fine-tuning.	Neural training required control of sequence length, numerical scale, and GPU workload. Text was limited to 512 tokens, targets were standardised for BamiBERT, and mixed precision was configured.
10 2–8 Nov.	Consolidated neural-model predictions, compared structured and text-based approaches, and developed nonnegative prediction blends with validation-fitted coefficients.	Combining separately trained models required aligned observations and consistent prediction units. Validation/test predictions retained common row identities and log-price units. Separate reconstruction of text-model ordering remained a consistency check.
11 9–11 Nov.	Brought together experiment summaries, diagnostic figures, and report contributions. Reviewed model settings, interpretation, and consistency between individual and combined predictions.	Results from different experimental populations could not be pooled directly. Comparisons were restricted to compatible runs; remaining text-filtering and checkpoint-compatibility limitations were retained for discussion.

Fortnightly tracking and discussion. Following the kickoff on 31 August, the tracking milestones were set two weeks apart: 14 September (data loading and quality issues), 28 September (cleaning decisions and feature preparation), 12 October (baselines and evaluation protocol), 26 October (tree/text results and neural-training requirements), and 9 November (prediction alignment, model combinations, and report integration). At each checkpoint, members reviewed deliverables, discussed unresolved issues, and agreed on the next tasks. A final review on 11 November covered the consolidated findings and outstanding limitations. Intermediate outputs were also exchanged between meetings so that modelling work could proceed in parallel.

4 Exploratory Data Analysis (EDA)

This section presents the data quality checks, the construction of the analytical dataset, and the price analysis. The work was carried out in two notebooks: the first covers data quality and cleaning, and the second covers the analysis of price and area. Each listing is treated as a property for sale, and `price` is the asking price in VND. The results therefore cover only listings that show a price, and some rental listings may remain (see Section 2).

4.1 Data Quality Analysis

4.1.1 Schema and Data Types

The raw data contain 3,500,744 listings and the 19 variables described in Table 2. The schema check confirmed that all expected columns are present and that there are no additional columns. The data types are as follows:

- Numeric (8 variables): `price`, `area`, `floor_count`, `frontage_width`, `house_depth`, `road_width`, `bedroom_count`, `bathroom_count`.
- Text (11 variables): all remaining variables.

To reduce memory usage, six categorical columns were converted to the categorical type, which reduced the memory footprint from 6.533 GB to 5.334 GB. A conservative cleaning strategy was adopted: quality issues were flagged and all rows were kept in the full dataset, while a separate subset of priced listings was built afterwards (Section 4.2).

4.1.2 Missing-Value Analysis

The null-value scan showed missing entries in 14 of the 19 variables (Table 4). The main findings are:

- `price`: 218,494 missing values (6.24%), and `area`: 4 missing values.
- Structural attributes have high missing rates, ranging from 68.96% for `house_direction` to 98.04% for `house_depth`.
- `name`, `description`, `property_type_name`, `province_name`, and `published_at` have no missing values.

Table 4: Missing values by variable in the raw data ($N = 3{,}500{,}744$).

Variable	Missing (n)	Missing (%)
`house_depth`	3,432,288	98.04
`balcony_direction`	2,933,342	83.79
`road_width`	2,830,873	80.86
`floor_count`	2,691,982	76.90
`project_name`	2,499,891	71.41
`house_direction`	2,414,214	68.96
`frontage_width`	1,726,424	49.32
`bathroom_count`	1,720,974	49.16
`bedroom_count`	1,600,361	45.71
`street_name`	1,376,860	39.33
`ward_name`	496,716	14.19
`price`	218,494	6.24
`district_name`	101,933	2.91
`area`	4	<0.01

The missing rate was also computed for each property type.

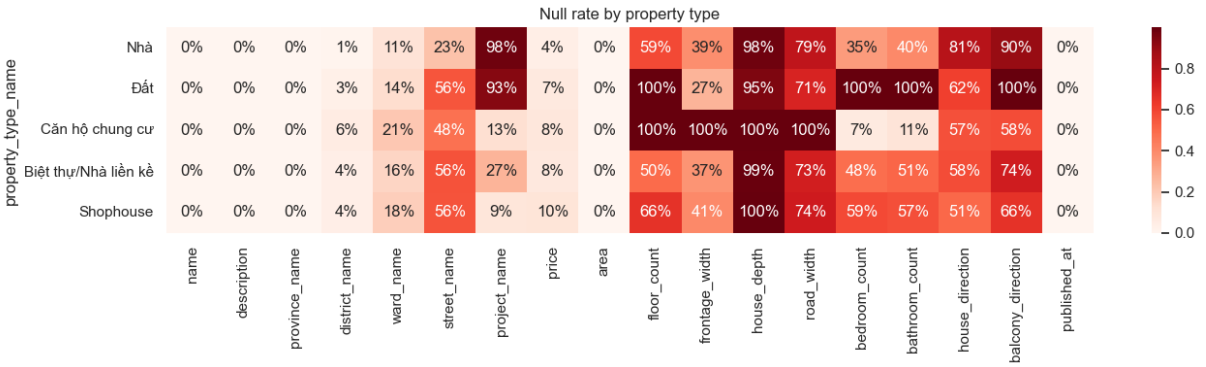


Figure 1: Null rate of each variable by property type.

Figure 1 shows the null rate of each variable for the five property types. The pattern differs across types, which indicates that part of the missingness is related to the nature of the property:

- Apartments (*Căn hộ chung cư*): `floor_count`, `frontage_width`, `house_depth`, and `road_width` are missing in 100% of the listings.
- Land (*Đất*): `floor_count`, `bedroom_count`, `bathroom_count`, and `balcony_direction` are missing in 100% of the listings.
- `project_name` is missing in 98% of houses (*Nhà*) and 93% of land listings, compared with 13% of apartments. A missing `project_name` often indicates that the listing does not belong to a named development project.
- `price` is missing in 4% to 10% of the listings of each type.

Rows with missing values were retained in the dataset, and listings without a valid price are excluded later from the price analysis.

4.1.3 Validity Checks

Validity checks were applied to the numeric variables:

- `price`: 57,320 listings have a price equal to zero, and no listing has a negative price.
- `area`: no listing has an area less than or equal to zero.
- `floor_count`, `bedroom_count`, `bathroom_count`: no negative values.

Listings without a valid price or area represent 7.88% of the raw data. This share ranges from 5.2% (*Nhà*) to 13.6% (*Shophouse*) across the five property types, and from 4.1% (Khánh Hòa and Lâm Đồng) to 18.8% (Hòa Bình) among the 15 provinces with the most listings. These listings are therefore spread across property types and provinces, and no group accounts for most of them.

4.1.4 Temporal, Geographic and Direction Checks

The following checks were applied to the time, location, and direction variables:

- `published_at`: no missing values in the raw data, but 22,024 values could not be converted to a timestamp (invalid values were coerced to missing). The parsed values span 1 June 2025 to 30 March 2026.
- Geography: the data contain 63 provinces and 627 districts. A total of 15 district names appear in more than one province, so `district_name` should be used together with `province_name`.
- Direction: `house_direction` and `balcony_direction` are free text, with 2,414,214 and 2,933,342 missing values, respectively. The same direction is sometimes written in two notations (for example, *Đông Nam* and *Đông - Nam*), and a few entries contain several directions, each occurring at most five times in `house_direction` and at most four times in `balcony_direction`. No standardisation was applied at this stage.

4.1.5 Duplicate Analysis

Listings were flagged as duplicates when they share the same normalised text of `name`, `province_name`, `district_name`, and `ward_name` (lower case, diacritics removed, *đ* mapped to *d*, whitespace collapsed), together with the same `area` and `price`, as an earlier listing. The first occurrence is kept.

```
df["is_dup"] = dup_check.duplicated(keep="first")
```

Under this definition, 865,362 listings (24.72% of the raw data) are duplicates. The ten most repeated combinations occur between 60 and 121 times each, all in Hồ Chí Minh, and eight of the ten share the same title opening (*xuat canh ban gap*, in normalised form). The most repeated combination is a house on Lê Thánh Tôn street (Bến Nghé ward, District 1) of about 43 m^2, priced at 15.5 billion VND, which appears 121 times.

4.1.6 Extreme Value Analysis

The distribution of `price` in the priced, de-duplicated data was plotted on logarithmic axes.

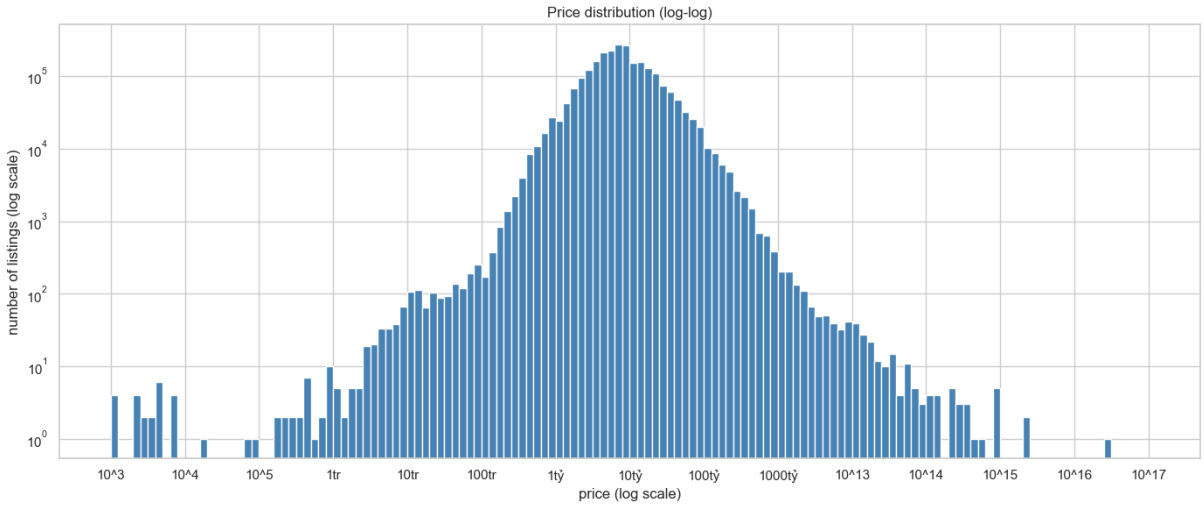


Figure 2: Distribution of asking price after removing invalid prices and duplicates (logarithmic scale on both axes).

Figure 2 shows that most listings are concentrated between about 10^9 and 10^{10} VND, with isolated listings below 10^4 VND and above 10^{15} VND. The ten lowest and ten highest prices were inspected:

- The ten lowest prices are between 1,000 and 3,000 VND. In several of these listings the title states a different price; for example, a listing recorded at 2,000 VND is titled as 1.6 billion VND.
- The ten highest prices range from about 4.49×10^{14} to 2.66×10^{16} VND, and some of these listings have very large areas (for example, 999,999 m^2).

These observations suggest data entry errors, for example in the unit of the price. A range of $[10^8, 10^{12}]$ VND (100 million to 1,000 billion VND) was therefore examined: it would remove 2,574 listings (0.107%), and it was applied as a cleaning step (Section 4.2).

The numeric attributes and the price per square metre were inspected in the same way.

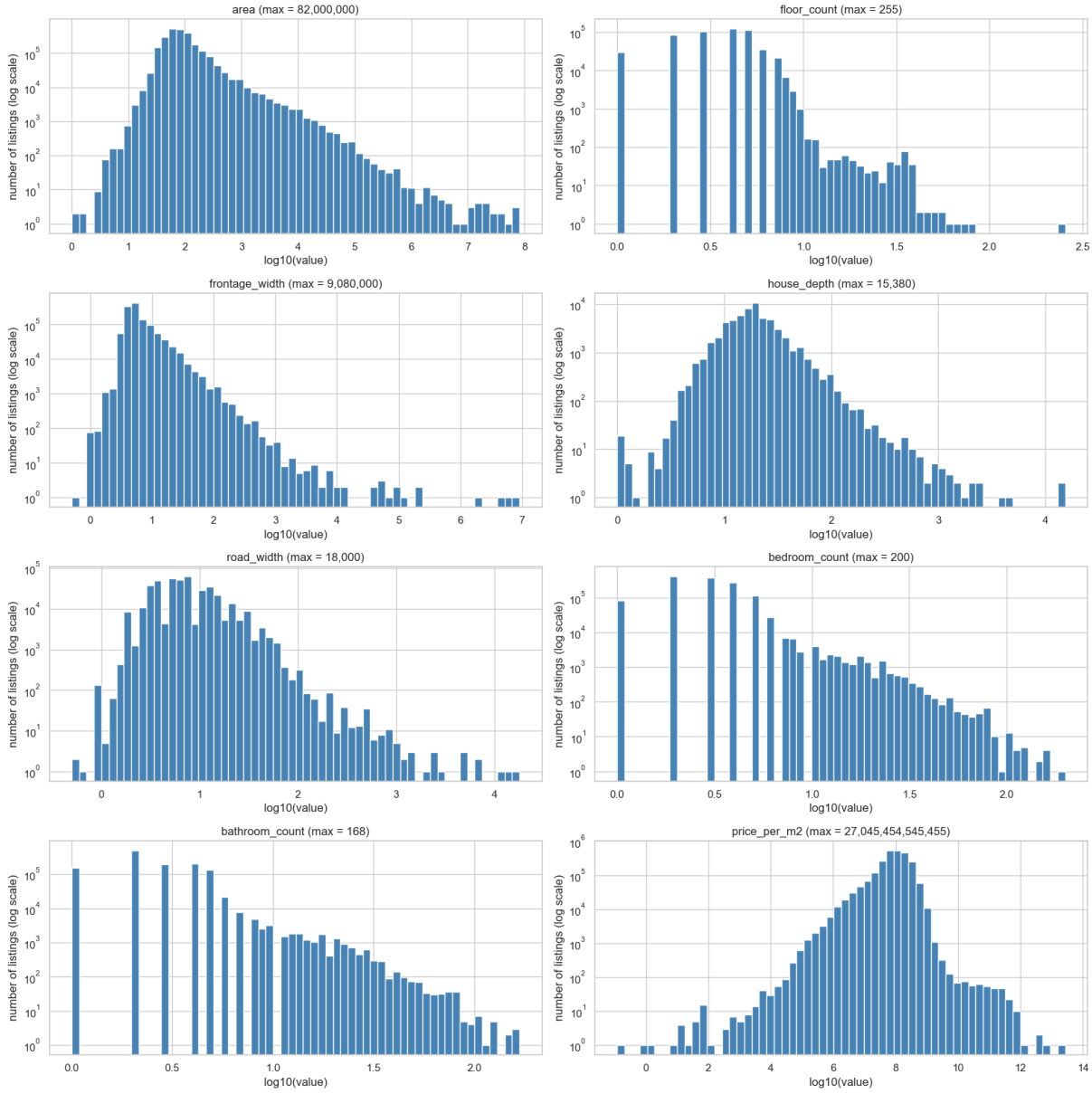


Figure 3: Distributions of the numeric attributes and of the price per square metre ($\log_{10}$ scale; vertical axis in logarithmic scale), before the plausibility limits are applied.

Figure 3 shows that in each panel most listings lie in a narrow range, while sparse tails extend over several orders of magnitude. The maximum values are 82,000,000 m^2 for `area`, 255 for `floor_count`, 9,080,000 m for `frontage_width`, 15,380 m for `house_depth`, 18,000 m for `road_width`, 200 for `bedroom_count`, 168 for `bathroom_count`, and about 2.70×10^{13} VND per m^2 for `price_per_m2`. These values are not plausible for most properties. The separated bars for `floor_count`, `bedroom_count`, and `bathroom_count` reflect that these variables take integer values.

4.2 Analytical Dataset Construction

The analytical dataset (`df_priced`) was built from the full dataset in six steps, applied in the order of Table 6. The first three steps retain listings with a valid price and area, remove the flagged duplicates, and retain prices within $[10^8, 10^{12}]$ VND:

```
df_priced = df[df["price"].gt(0) & df["area"].gt(0)]
df_priced = df_priced[~df.loc[df_priced.index, "is_dup"]]
df_priced = df_priced[df_priced["price"].between(1e8, 1e12)]
```

The remaining steps remove listings without a valid publication date, apply plausibility limits to `area` and the other numeric attributes (Table 5), and remove listings with a price per square metre above 2 billion VND. For `area`, listings outside the limits were removed. For the other six attributes, the out-of-range values were set to missing and the listings were retained. The number of values outside the limits was counted before the area filter was applied. The final dataset contains 2,397,001 listings.

Table 5: Plausibility limits applied to the numeric attributes.

Variable	Valid range	Values outside (n)	Treatment
`area` (m^2)	$[10, 100{,}000]$	876	Row removed
`floor_count`	$[1, 50]$	8	Set to missing
`frontage_width` (m)	$[1, 200]$	761	Set to missing
`house_depth` (m)	$[2, 500]$	70	Set to missing
`road_width` (m)	$[1, 100]$	424	Set to missing
`bedroom_count`	$[1, 50]$	340	Set to missing
`bathroom_count`	$[1, 50]$	267	Set to missing

Table 6: Number of listings after each filtering step.

Step	Rule	Listings (n)	Removed (n)
0	Raw data	3,500,744	—
1	Price > 0 and area > 0	3,224,930	275,814
2	Duplicates removed	2,415,372	809,558
3	Price within $[10^8, 10^{12}]$ VND	2,412,798	2,574
4	Valid publication date	2,398,343	14,455
5	Area within $[10, 100{,}000]$ m^2	2,397,467	876
6	Price per $m^2 \leq 2 \times 10^9$ VND (final)	2,397,001	466

The number removed in step 2 is the difference between the counts before and after the step.

4.3 Price Analysis

All analyses in this subsection use the analytical dataset (`df_priced`).

4.3.1 General Dataset Profile

Table 7 reports descriptive statistics of the numeric variables. The main observations are:

- `price` and `area` are available for all listings, whereas the other attributes are available for fewer listings (from 58,878 for `house_depth` to 1,304,397 for `bedroom_count`).
- The median price is 7.4 billion VND and the mean (1.49×10^{10} VND) is about twice as large, which indicates a right-skewed distribution.
- `area` behaves similarly: the median is 80 m^2, the mean is 217 m^2, and the standard deviation is 1,619 m^2.
- The maximum of each attribute equals the upper limit in Table 5 (for example, 100,000 m^2 for `area` and 200 m for `frontage_width`), so the tails are bounded.

4.3.2 Distribution of Price and Area

The distributions of `price` and `area` were plotted after a base-10 logarithmic transformation.

Table 7: Descriptive statistics of numeric variables in the analytical dataset ($N = 2{,}397{,}001$).

Variable	Count	Mean	Std. dev.	Min	25%	50%	75%	Max
price (VND)	2,397,001	1.49×10^{10}	3.20×10^{10}	1.00×10^{8}	4.05×10^{9}	7.40×10^{9}	1.43×10^{10}	1.00×10^{12}
area (m^2)	2,397,001	217.38	1,619.32	10.00	56.00	80.00	120.00	100,000
floor_count	516,628	3.90	1.76	1	3	4	5	50
frontage_width (m)	1,208,300	7.54	8.97	1.00	4.20	5.00	7.50	200
house_depth (m)	58,878	22.57	20.84	2.00	13.41	18.75	24.64	500
road_width (m)	414,645	9.08	8.62	1.00	4.00	6.00	10.50	100
bedroom_count	1,304,397	3.30	2.34	1	2	3	4	50
bathroom_count	1,220,972	3.05	2.34	1	2	2	4	50

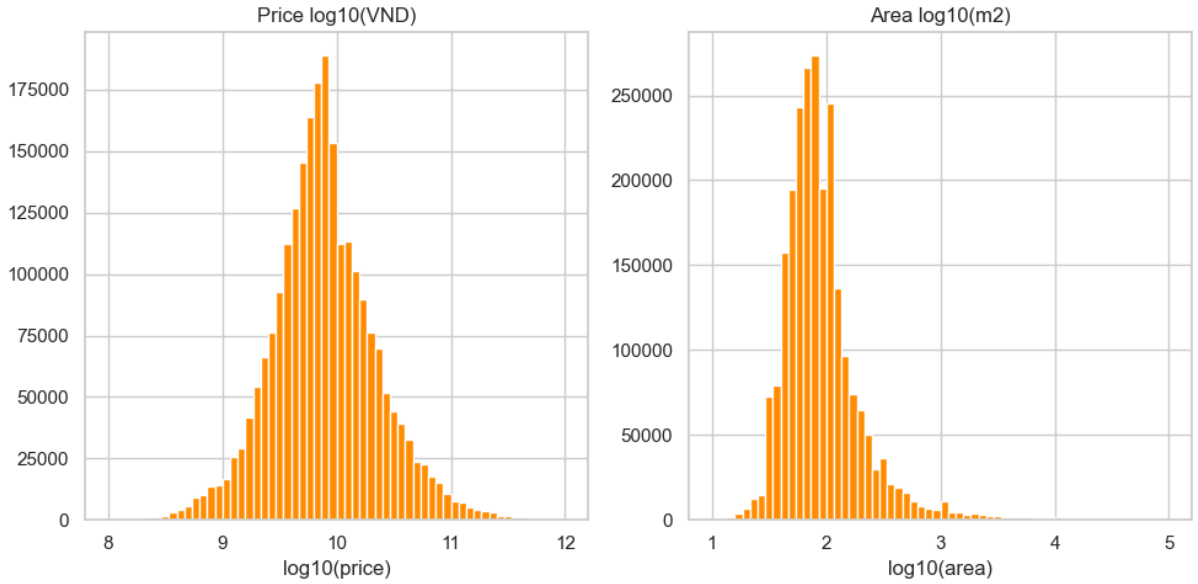


Figure 4: Histograms of $\log_{10}$(price) (left) and $\log_{10}$(area) (right).

Figure 4 shows that the distribution of price on the logarithmic scale is unimodal and approximately symmetric, with a mode at about 9.9 (roughly 8 billion VND) and values mainly between 8.5 and 11.5. The distribution of area is unimodal with a mode at about 1.9 to 2.0 (roughly 80 to 100 m^2) and a right tail that extends to about 5 (the upper limit of 100,000 m^2). Given the right skew shown in Table 7, this supports the use of a log transformation for price.

4.3.3 Group Coverage

The number of listings in each property type and province was counted. Groups with fewer than 100 listings are hard to use as features.

- Property types: all 5 types have more than 100 listings: houses (1,074,788), land (560,294), apartments (544,268), villas/townhouses (178,654), and shophouses (38,997).
- Provinces: 60 of the 63 provinces have at least 100 listings.
- Listings are concentrated in Hồ Chí Minh (928,687) and Hà Nội (776,760), which together account for about 71% of the analytical dataset.

4.3.4 Area–Price Relationship

The relationship between area and price was examined with a scatter plot on logarithmic axes, using a random sample of 20,000 listings (random seed 42) to limit overplotting.

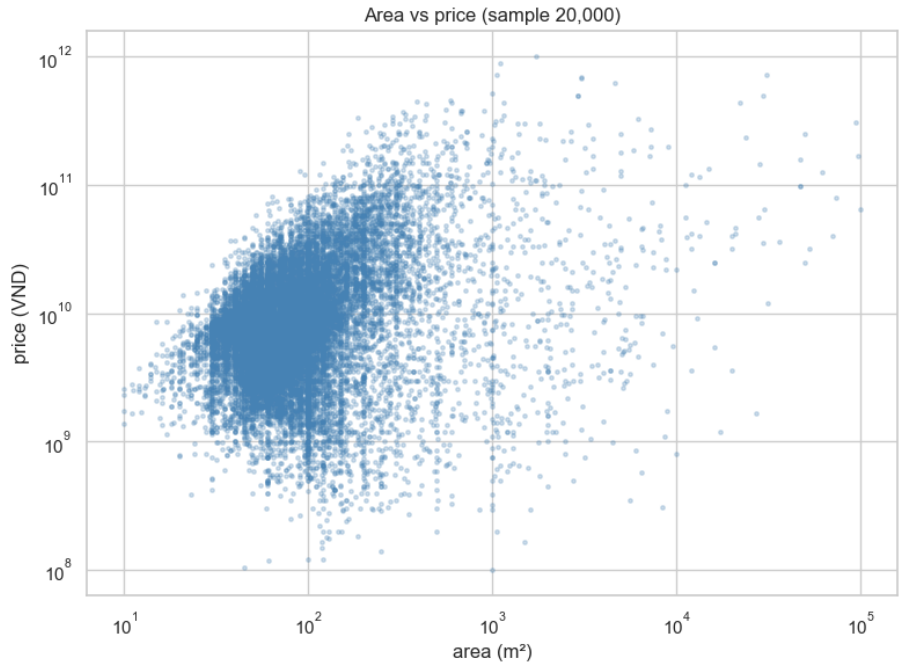


Figure 5: Area versus asking price on logarithmic axes (random sample of 20,000 listings).

Figure 5 shows that the area axis spans 10 to 10^5 m^2, the limits applied in the cleaning. Most points lie at areas of roughly 30 to 300 m^2 and prices of roughly 10^9 to 10^{10} VND. The cloud shows an upward tendency, so larger areas are generally associated with higher prices, but the dispersion at a given area is wide, and points become sparse above 10^3 m^2. This plot is a visual indication only; the strength of the association was not quantified.

4.3.5 Price per Square Metre by Property Type

The price per square metre (`price_per_m2`) is `price` divided by `area`. Its $\log_{10}$ distribution was plotted for each of the five property types, with the horizontal axis restricted to 6–10. A property type whose histogram has two peaks may mix different kinds of listing (for example, sale and rent).

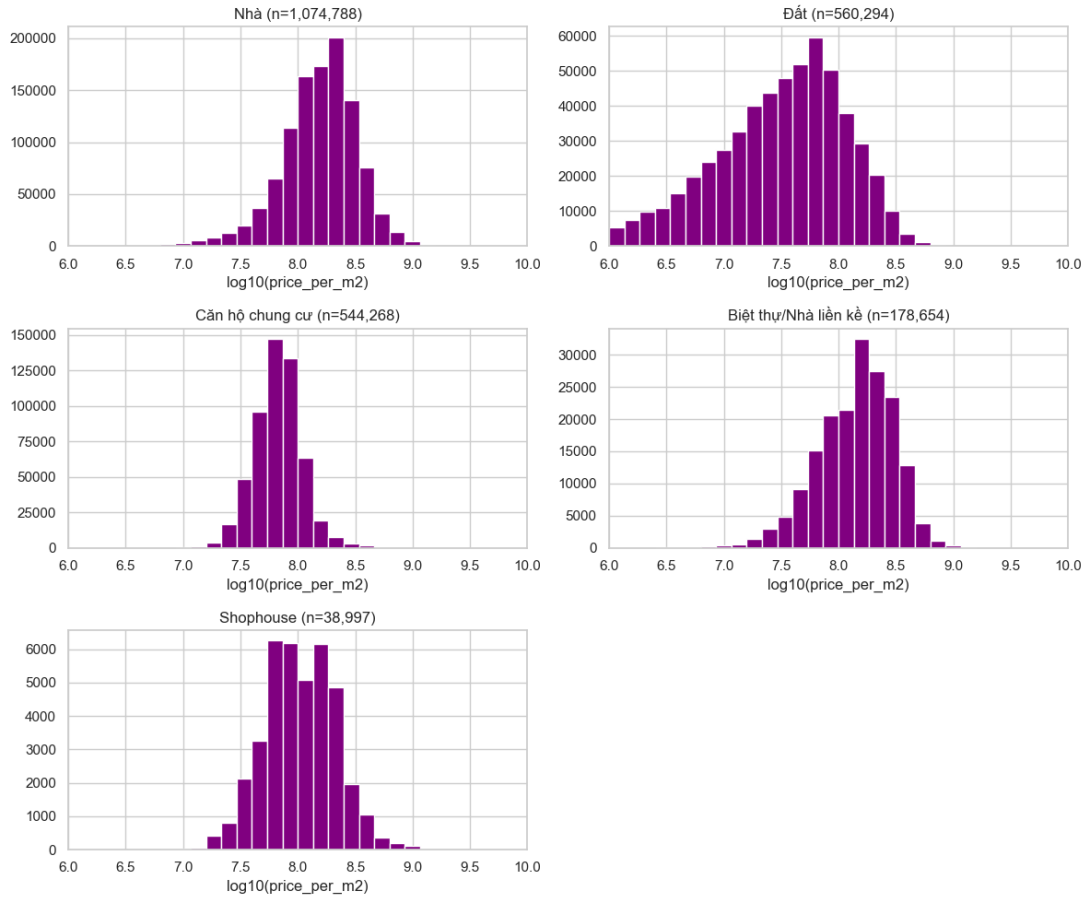


Figure 6: Distribution of $\log_{10}$(price per m^2) by property type (houses $n = 1{,}074{,}788$; land $n = 560{,}294$; apartments $n = 544{,}268$; villas/townhouses $n = 178{,}654$; shophouses $n = 38{,}997$).

Figure 6 shows that the distributions differ in location and spread across property types (the data are capped at 2 billion VND per m^2, i.e. $\log_{10} \approx 9.3$):

- Houses and villas/townhouses are centred at higher values (modes at about 8.3 and 8.2, roughly 200 and 160 million VND per m^2) than apartments and land (modes at about 7.8).
- Land has the widest distribution, skewed to the left, and apartments have the narrowest.
- Shophouses show two local maxima, at about 7.8–7.9 and 8.2. This may indicate a mix of different kinds of listing (for example, sale and rental), but it was not tested and remains a hypothesis.

4.4 Summary of Findings

- The raw data contain 3,500,744 listings. Missing values are frequent for structural attributes and depend on the property type.
- 24.72% of the raw listings were flagged as duplicates and 7.88% lack a valid price or area. After all filtering steps, 2,397,001 listings remain.
- Price and area are right-skewed, and the log transformation of price gives an approximately symmetric distribution. Extreme values were handled with fixed plausibility limits.
- All five property types and 60 of 63 provinces have sufficient listings, although listings are concentrated in two provinces. Area and price show a positive but widely dispersed relationship, and the price per square metre differs by property type.

5 Model Selection and Training

5.1 Machine Learning

The machine learning experiments predict property asking prices from physical attributes, location, property category, and advertising text. The experimental design progresses from constant and local-median baselines to regularised linear regression, tree ensembles, and models combining textual and structured information. This progression assesses whether nonlinear interactions and listing descriptions contribute predictive information beyond a linear representation of property attributes.

Figure 7 summarises the modelling workflow. All models share the same chronological training, validation, and test partitions. Learned preprocessing is fitted on training observations during model selection; subsequent refitting procedures are specified separately for each model family.

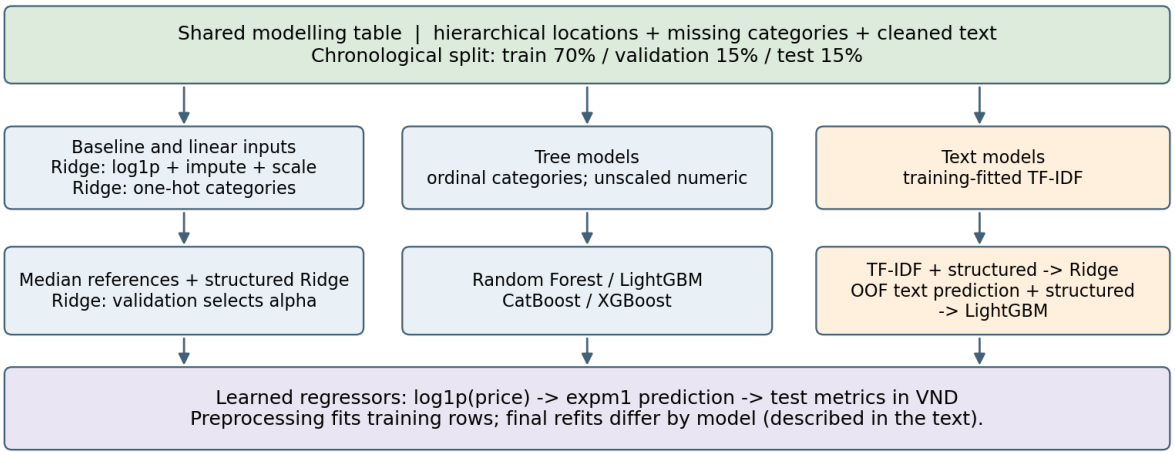


Figure 7: Machine learning workflow from shared feature preparation to model fitting and evaluation. Median baselines predict prices directly, while the learned regressors predict log price. OOF denotes out-of-fold predictions used to construct the training input for the text-enhanced LightGBM.

5.1.1 Feature Representation and Training Protocol

Structured predictors. The structured representation contains seven numeric and eight categorical variables (Table 8). The target is the asking price in Vietnamese dong (VND). Publication time determines the chronological partitions and is excluded from the predictor matrix. A listing's own price per square metre and other target-derived quantities are also excluded from ordinary predictors.

Table 8: Structured features used by the machine learning models.

Feature group	Variables
Numeric	Area, floor count, frontage width, house depth, road width, bedroom count, bathroom count
Categorical	Property type, province, district, ward, street, development project, house direction, balcony direction

Location categories are represented hierarchically: district names are prefixed with their province, while ward and street names are prefixed with the resulting district key. This distinguishes identical local names in different administrative areas. Missing categorical values receive a dedicated category. Numeric missing values are retained for model-specific treatment.

Text representation. The title and description are concatenated, lowercased, and stripped of the phone-number placeholder. To reduce direct target disclosure, punctuation-delimited clauses containing monetary terms, including *giá*, *tỷ*, *triệu*, VND, and USD, are removed. Tokens containing digits are subsequently removed, and whitespace is normalised. This conservative processing also removes useful

numeric descriptions, such as area and room counts; these remain available through the structured predictors. The monetary-clause filtering is heuristic and cannot guarantee complete removal of every price expression.

Chronological evaluation. The 2,397,001 prepared listings are ordered by publication time. The oldest 70% form the training partition (1,677,900 observations), the next 15% form validation (359,550), and the most recent 15% form testing (359,551). This evaluates prediction on later listings using earlier observations. The test partition is excluded from fitting preprocessing transformations, selecting hyperparameters, and determining stopping points.

Target transformation. The median baselines predict original-scale prices. All learned regressors use the natural-log target

$$z_i = \log(1 + y_i), \qquad \widehat{y}_i = \exp(\widehat{z}_i) - 1, \tag{1}$$

where y_i denotes asking price in VND. The transformation compresses the upper price tail during fitting. Predictions are transformed back before calculating original-scale evaluation metrics. Because log-price loss and price loss weight errors differently, validation criteria are identified explicitly below.

5.1.2 Baseline Models and Ridge Regression

Median-price baseline. The first baseline assigns every property the median training price:

$$\widehat{y}_i = \text{median}_{j \in \mathcal{T}}(y_j), \tag{2}$$

where $\mathcal{T}$ denotes the training partition. This constant predictor provides an absolute-error benchmark without using property attributes.

Local median price per square metre. The second baseline uses area, district, and property type. For each district–property-type group g, the median price per square metre is estimated from training observations:

$$m_g = \text{median}_{j \in \mathcal{T}: g_j = g} \left(\frac{y_j}{a_j} \right), \qquad \widehat{y}_i = a_i m_{g_i}, \tag{3}$$

where a_i is area. A group absent from training uses the overall training median price per square metre. This is a study-specific reference model for local area-based pricing; no validation or test targets enter its lookup values.

Ridge regression. Ridge regression introduces an L_2 penalty to stabilise linear coefficient estimation [HK70]. It serves as the regularised linear benchmark for the structured predictors. Numeric variables undergo a logarithmic transformation, median imputation, addition of missingness indicators, and standardisation. Categorical variables are one-hot encoded, pooling levels observed fewer than 50 times. Unseen categories use the pooled category when available. Candidate preprocessing statistics are estimated exclusively from training rows.

The regression objective is

$$\min_{b, \boldsymbol{\beta}} \sum_{i \in \mathcal{T}} \left(z_i - b - \mathbf{x}_i^\top \boldsymbol{\beta} \right)^2 + \alpha \|\boldsymbol{\beta}\|_2^2, \tag{4}$$

where α controls coefficient shrinkage [HK70]. Three values, $\alpha \in \{1, 10, 100\}$, are compared using validation MAE after inversion to VND. The selected value is $\alpha = 1$. The entire preprocessing and regression pipeline is subsequently refitted on training plus validation observations before test prediction.

Figure 8 illustrates the fitted model's test predictions. Points close to the diagonal indicate agreement between actual and predicted prices. Several expensive properties fall below the diagonal, indicating underprediction in the upper price range.

5.1.3 Tree Ensemble Models

Random Forest aggregates randomised regression trees [Bre01], while gradient boosting constructs an additive predictor through successive loss-reducing updates [Fri01]. These complementary ensemble approaches allow nonlinear relationships and interactions between physical attributes, property type, and location.

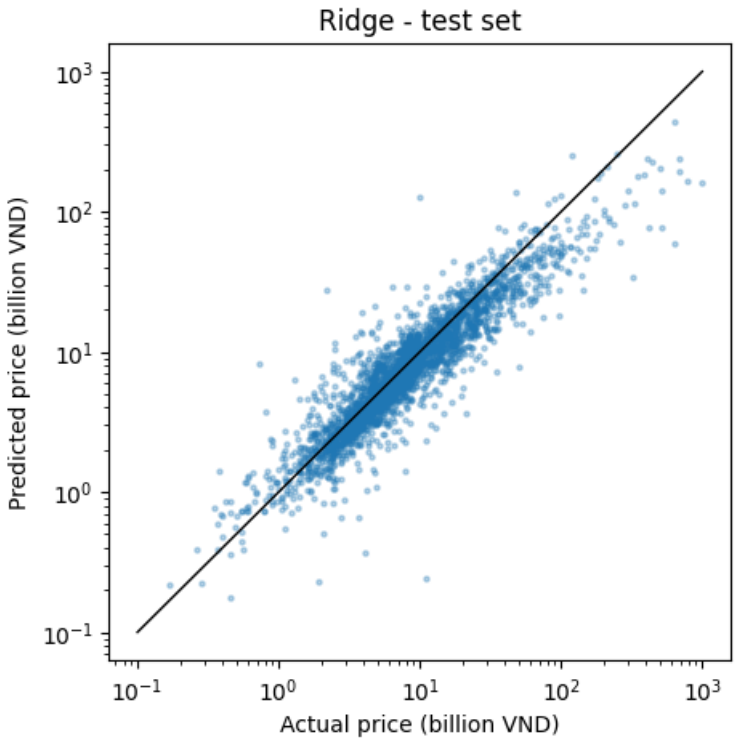


Figure 8: Actual and predicted asking prices for a random sample of 3,000 test listings (seed 42) using structured Ridge regression. Both axes are logarithmic and expressed in billion VND. The diagonal denotes exact prediction.

Input encoding. Categorical levels are mapped to integer codes using a training-fitted encoder; unseen levels receive code -1. Numeric features retain their original scale and missing values. Random Forest treats category codes as numeric inputs. LightGBM and CatBoost receive explicit categorical-column declarations, while XGBoost receives categorical-typed columns with categorical support enabled. Consequently, the integer encoding does not imply the same treatment of categories across all four models.

Random Forest. Following the randomised-tree aggregation framework of Breiman [Bre01], the forest contains 100 regression trees, with at least five observations per leaf and random seed 42. Each tree draws a bootstrap sample containing 10% of the fitting population to reduce the training cost per tree.

The search crosses maximum depth $\{12, 20\}$ with feature fraction $\{0.5, 1.0\}$, producing four candidate configurations. Candidates fit training log prices and are ranked by validation MAE in VND. Maximum depth 20 and feature fraction 1.0 are selected. The final forest is refitted on training plus validation observations, while the categorical mapping remains fitted on training data.

LightGBM. LightGBM is a scalable gradient-boosted decision-tree framework [Ke+17]. The model uses up to 2,000 trees, learning rate 0.05, 63 leaves, and seed 42. Validation log-MAE and the default log-MSE are monitored, with early-stopping patience of 50 rounds. Categorical columns are declared explicitly, and numerical features are not standardised.

Figure 9 reports gain-based importance, a tree-model interpretation based on accumulated loss reduction from splits [Fri01]. District and area have the highest gains. This indicates their importance to the fitted predictor, rather than a causal effect or an independent contribution in the presence of correlated variables.

CatBoost. CatBoost provides gradient boosting with specialised treatment of categorical features [Pro+18]. The fitted configuration allows 2,000 iterations with learning rate 0.1, maximum categorical-combination complexity one, 64 numerical quantisation borders, and seed 42. Categorical columns are explicitly identified in the integer-encoded feature table. The training loss is RMSE on

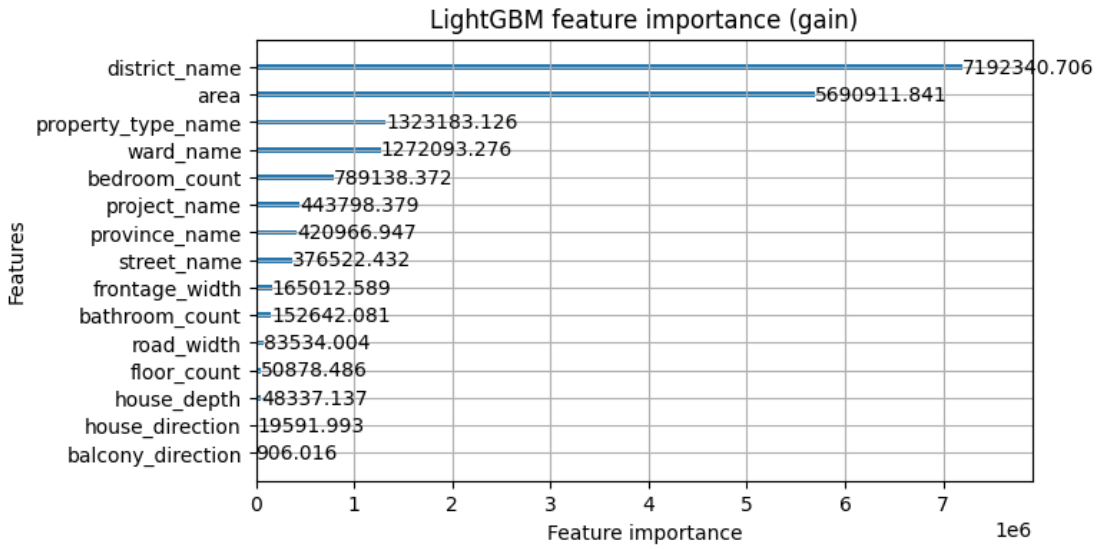


Figure 9: LightGBM feature importance measured by accumulated split gain. District categories incorporate the province prefix. Gain values are model-dependent quantities, rather than percentages or effects in VND.

log price, while validation log-MAE determines checkpoint selection with early-stopping patience of 30 rounds.

XGBoost. XGBoost is a scalable tree-boosting system developed by Chen and Guestrin [CG16]. The configuration uses up to 2,000 trees, learning rate 0.05, maximum depth eight, histogram-based tree construction, categorical support, parallel CPU execution, and seed 42. Training operates on log price, monitors validation log-MAE, and uses early-stopping patience of 50 rounds.

The boosting configurations are fixed experimental settings rather than the outcome of a parameter-grid search. All three reach the 2,000-round limit in the recorded training runs. Early stopping is therefore configured but does not shorten those runs. These models retain their training-fitted, validation-monitored predictors rather than undergoing a subsequent training-plus-validation refit.

Table 9: Training configurations and selection procedures for structured regressors.

Model	Main settings	Selection and final fitting
Ridge	$\alpha \in \{1, 10, 100\}$	Validation MAE in VND; $\alpha = 1$; refit on train + validation
Random Forest	100 trees; depth $\{12, 20\}$; feature fraction $\{0.5, 1.0\}$; leaf size 5	Validation MAE in VND; depth 20 and fraction 1.0; refit on train + validation
LightGBM	2,000 trees; learning rate 0.05; 63 leaves	Validation log-price metrics; patience 50; training-fitted model
CatBoost	2,000 iterations; learning rate 0.1; categorical-combination complexity 1; 64 borders	Validation log-MAE; patience 30; training-fitted model
XGBoost	2,000 trees; learning rate 0.05; depth 8; histogram method	Validation log-MAE; patience 50; training-fitted model

5.1.4 Combining Text and Structured Features

Textual information is incorporated using two complementary designs: a joint sparse representation fitted by Ridge, and a text-derived prediction used as an additional LightGBM input. Both preserve the structured attributes while introducing information expressed in the listing description.

TF-IDF representation. Term frequency–inverse document frequency (TF-IDF) assigns weights according to within-document term occurrence and corpus-level document frequency, following established term-weighting methods [SB88]. The representation uses word unigrams, minimum document frequency five, a maximum vocabulary of 50,000 terms, sublinear term frequency, and 32-bit floating-point storage. The resulting vocabulary contains 23,233 terms. Vocabulary selection and IDF estimation use training documents only; validation and test documents reuse the fitted representation.

Ridge on combined features. The TF-IDF matrix is horizontally concatenated with the preprocessed structured matrix, yielding 29,592 columns. Sparse storage avoids materialising a large dense design matrix. Ridge regularisation [HK70] is selected from $\{1, 10, 100\}$ using validation MAE in VND, with $\alpha = 1$ selected. The final regression coefficients are fitted on the combined training and validation matrices. The vocabulary and structured preprocessing remain fitted on training observations, in contrast to the full-pipeline refit used for structured-only Ridge.

Figure 10 visualises the validation searches. Both Ridge variants select $\alpha = 1$, and the joint representation has lower validation MAE at all three tested values. The Random Forest comparison supports maximum depth 20 and feature fraction 1.0. These are model-selection results, rather than test-set performance estimates.

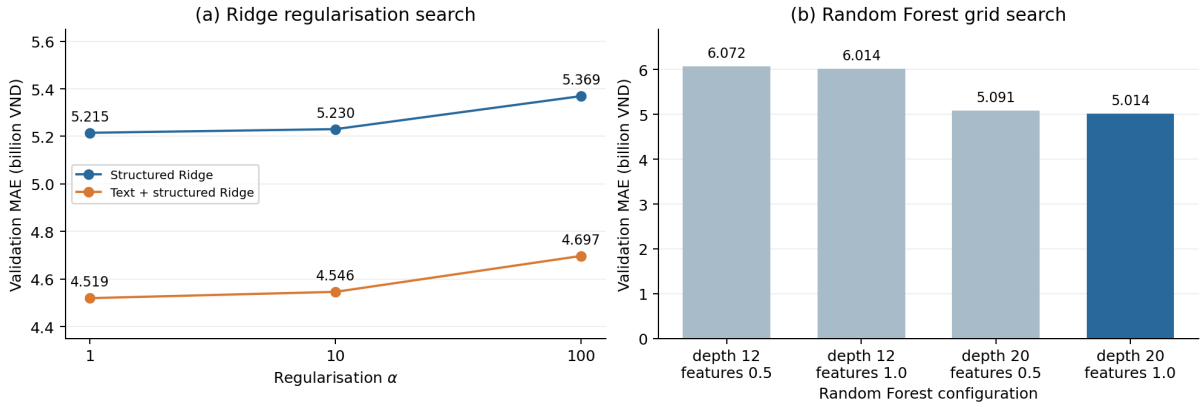


Figure 10: Validation-based hyperparameter selection: (a) structured Ridge and Ridge with text and structured features; (b) the four Random Forest configurations. Lower MAE is better. The darker bar indicates the selected forest; values are rounded to three decimals.

LightGBM with a text-derived prediction. The second design follows the prediction-as-feature principle of stacked generalisation [Wol92]. A text-only Ridge model predicts log price, and that prediction is appended to the 15 structured attributes as a sixteenth input to LightGBM. To avoid using in-sample fitted target estimates as training features, five-fold cross-validation produces out-of-fold text predictions: each fold is predicted by a Ridge model fitted on the other four folds.

A text Ridge model is then fitted on all training documents to produce validation and test features. Its regularisation parameter reuses the value selected for the joint Ridge model. LightGBM [Ke+17] is trained on the augmented feature matrix with the same 2,000-tree limit, learning rate 0.05, 63 leaves, seed 42, and early-stopping patience of 50 rounds as the structured model.

The internal five-fold procedure does not use forward-chaining splits, and the vocabulary is estimated from the complete outer training partition before fold-wise Ridge fitting. Thus, each out-of-fold prediction excludes its own target from Ridge coefficient estimation, but the internal procedure does not reproduce the chronological outer evaluation at every preprocessing stage. This limitation is retained when interpreting the text-enhanced model.

5.1.5 Evaluation and Prediction Protocol

All original-scale predictions are evaluated using MAE, RMSE, R^2, and MAPE. MAE and RMSE are expressed in billion VND, while MAPE is expressed as a percentage. MAE measures typical absolute price

error, RMSE emphasises large errors, R^2 compares squared error with a constant mean-price reference, and MAPE measures relative error. Positive target prices ensure nonzero MAPE denominators.

For subsequent model combination, validation and test predictions from the boosting and text-enhanced models are retained in log-price units with a common listing identity. The combined-feature Ridge supplies validation predictions from the selected training-only model and test predictions from the model refitted on training plus validation. This prevents the validation estimates from being generated by a regressor already fitted on their targets. Boosting predictions come from their training-fitted, validation-monitored models.

The common outer partition supports comparison across model families, although their final fitting populations differ because only some models are refitted on training plus validation. The heuristic removal of monetary text and the internal cross-validation procedure are additional limitations when interpreting the contribution of textual information.

5.2 Deep Learning

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